

Internal Representation Equivalences

One generative core, three relationship types: identity, bijection, projection — the
Doc. 265 operator turned inward

Johann Pascher

ORCID: 0009-0000-6518-4064

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Abstract

FFGFT exists in several mathematical languages – torus geometry, matrix algebra, Hilbert-space operators, information formalism, recursion, two layers. This document does not ask how FFGFT stands relative to *other* frameworks (Doc. 265 does that outward); it asks how these internal representations stand *among themselves*. Finding: they are not a flat set of equals but a *generative core* (the T^4 torus) plus three clearly distinct relationship types – **identity, bijection, projection**. The geometry and the 3×3 matrix algebra are *the same thing* (Doc. 206); the standard-QM Hilbert space is a *lossless bijection* (Doc. 230); the information formalism and the spatial reduction are *lossy projections* with a formalism-relative residue (Doc. 243/270). The decisive point: *only* the full T^4 geometry loses nothing. The $\theta=2/9$ work carries the same result from the other side: all representations are faithful on the radial (magnitude) sector; the only irreducible part is the angular (phase) sector.

.1 The question and its boundary

Over recent months FFGFT has been cast in strikingly many languages, and the impression of several “parallel roads” is justified. The question here is internal: *what relationship* holds between these roads? Are they interchangeable pictures of one object, or do they differ structurally?

This must be set apart from Doc. 265. There a *comparison operator* (three levels – near-equivalence, structural overlap, complementary; six axes) is developed and applied to FFGFT *against an external framework* (Blume-Maleev). Here the same operator is turned *inward*: not FFGFT versus foreign, but FFGFT-representation versus FFGFT-representation. The result is sharper than the outward operator allows – and it exposes a relationship type the symmetric outward operator does not have (Section .7).

.2 The generative core: why T^4 is not one representation among many

Two documents say the same thing from different sides. Doc. 152 (ontological priority) sets the T^4 torsion crystal as *fundamental ontological reality*; the fractal granulation $D_f = 3 - \xi$, the effective laws and classical 3D physics are emergent approximations “projected upward”. Doc. 243 gives the structural counterpart: every information formalism discards at least one of the three quantities θ_4 , (n_θ, n_φ) , ε_i ; the T^4 geometry is the *only* formalism that structurally holds all three at once.

Key Result

The T^4 torus is not one representation among many but the generative core: the only language that holds the full content. All other representations are maps *from* T^4 – and split, by the type of that map, into three classes.

.3 The three relationship types

Identity: geometry $\equiv$ matrix algebra (Doc. 206)

The Z_3 triangle (as a graph) and the 3×3 matrix A are not merely isomorphic but *algebraically identical*: the spectrum of A is the character table of Z_3 . Doc. 206 puts it literally: geometry

and matrix algebra are “not two pictures of the same thing, they are the same thing.” This is the strongest type – not a lossless translation channel but identity. Honestly noted: this identity is *sector-local*. It holds on the C_3/Z_3 core (3×3). Globally – with the angular phase θ and the φ harmonics – it is no longer a finite identity; the circulant captures the magnitude spectrum, not the phase (Doc. 286).

Bijection: the lossless channel (Doc. 230, Doc. 070)

The Hilbert-space translation (Doc. 230) is a *bijective structural statement*: every QM calculation sits inside the FFGFT formalism and back, *without loss*. It is no poorer than FFGFT and no richer in computation – but FFGFT *adds* what standard QM does not provide: the *why* of the constant values. In the projection-chain picture (Doc. 270) this is the *upward* move (Type III), and precisely because it introduces no geometric extra dimensions it is lossless.

A second bijection case is corpus-internal: the two mass forms in Doc. 070, the compact (cancelled) and the extended (geometrically original). They are mathematically equivalent, but only the extended carries the physical origin – which is why the ratios must not be cancelled directly.

Projection: the lossy, directed channel (Doc. 243, Doc. 270)

Doc. 243 maps FFGFT into an information formalism – as a *projection* π , *not* an isomorphism. A residue is lost, and that residue is *formalism-relative*: it depends on the chosen information formalism, not on the geometry. The same mapping into a neural network shares this character. Geometrically this corresponds to the *spatial* reductions of the projection chain (Doc. 270, Type II): strictly lossy, irreversible, with residue $\sim \xi$ (visible as a CHSH deviation).

To be set apart is the first reduction step $D4 \rightarrow D3$ (Doc. 270, Type I): not a lossy projection but a *duality decompactification* via $\tilde{T} \cdot m = 1$ – information-preserving; only the direct visibility of compactness is lost.

.4 Two axes that are not competing representations

Not every “language” is a rival encoding of the same content. Two are orthogonal separations:

- **Two layers** (Doc. 241): the separation of ratio-based, parameter-free content (Layer 1) from a single SI anchor (Layer 2). This is bookkeeping – what is input, what follows – not an alternative geometry. Orthogonal to the geometry/matrix/Hilbert choice.
- **Recursion / fixed point** (Doc. 203/275): the *dynamical* axis. It generates the hierarchy (the $\xi \rightarrow \varphi$ passage) and is – decisively – purely *real/radial*. It carries no angular (phase) generator.

.5 The decisive finding

Two statements that together form the strongest internal-consistency claim.

Important

(1) Only T^4 loses nothing. Every projective encoding drops a residue, and the residue is relative to the *representation*, not to the physics. The many roads therefore agree;

where they diverge it is exactly what the given encoding threw away. The "loss" is an artefact of the representation, not of the theory.

(2) The asymmetry (Doc. 270). Geometrically *downward* is lossy (Type II); representationally *upward* to the Hilbert space is lossless (Type III). This is exactly what lets FFGFT be *simultaneously* a "4D torus theory" and "ordinary QM" without contradiction.

.6 The bracket to the $\theta=2/9$ phase

The representation analysis and the θ elimination chain (Doc. 282/286) are one result from two sides. All representations are faithful on the *radial* (magnitude) sector: identity (206) and bijection (230) capture it completely – ξ , the hierarchy, the Koide amplitude $Q = 2/3 = r\sqrt{2}$, the mean $\langle f \rangle$. The *one* place where the geometry is irreducible is the *angular* (phase) sector: the offset $\theta = 2/9$. Only the full T^4 geometry holds it; the real recursion does not reach it, the circulant does not capture it. Faithful in magnitude, irreplaceable in phase.

.7 The 265 operator turned inward

Turning the Doc. 265 comparison operator inward shows two things. First, the internal relationships are *sharper*: where the outward operator reached only *near*-equivalence (different derivation routes, identical abstract structure), the internal bijections are *exact* (isomorphism, Doc. 230) and the matrix case is even *identity* (Doc. 206). Second, the inward application exposes a type the three symmetric levels of the outward operator do not have: the *directed projection* (243/270-II). The 265 levels compare two formalisms symmetrically; the internal projection is directed ($T^4 \rightarrow$ encoding) and carries a residue. This is no defect of the 265 operator but a finding: inward, a fourth, asymmetric relation is needed.

.8 Open edges

Declared honestly, not hidden:

- The **identity matrix** $\equiv$ **geometry** (206) is sector-local (3×3 , C_3). A global finite identity including θ and the φ harmonics does not exist; whether it *can* exist is open (Doc. 286).
- The **residue table** of the information formalisms (243) is a working stand, not a proof of minimality that T^4 is the only complete formalism.
- The **265 comparison operator** is itself open with three declared residues (axis completeness, weighting, independence); the fourth, directed relation found here is to be added.

Balance: one core, three relations

The parallel roads of FFGFT are not of equal rank. The T^4 torus is generative – the only representation that loses nothing. Everything else stands to it in exactly one of three relations: *identity* (matrix algebra, sector-local), *bijection* (Hilbert space; compact/extended mass form), *projection* (information formalism, neural network, spatial reduction – with a formalism-relative residue). Two further languages (two layers, recursion) are not rivals but orthogonal axes. The common ground of all roads is the radial magnitude sector; the only irreducible place is the angular phase $\theta = 2/9$.

Table 1: The internal representations, by relationship type to the generative T^4 core.

Representation	Doc.	Relationship	keeps / loses
Z_3 triangle $\equiv 3 \times 3$ matrix	206	identity	same thing; sector-local (C_3)
Hilbert space $H = L^2(T^4) \otimes \mathbb{C}^3$	230	bijection (loss-less)	QM $\leftrightarrow$ FFGFT; FFGFT adds the "why"
compact vs. extended mass form	070	bijection (equivalent)	equal; only extended carries origin
information formalism	243	projection π	residue, formalism-relative
neural network	243	projection	same
spatial reduction $T^3 \rightarrow T^0$	270	projection (Type II)	residue $\sim \xi$ (CHSH)
duality decompact. $D4 \rightarrow D3$	270	Type I (preserving)	only visibility of compactness
two layers (ratio/SI)	241	orthogonal axis	bookkeeping, no rival
recursion / fixed point	203/275	orthogonal axis	dynamical, purely real/radial

The representation question is thereby closed as far as it honestly can be: faithful in magnitude, generative only in the full torus, open only in the phase.